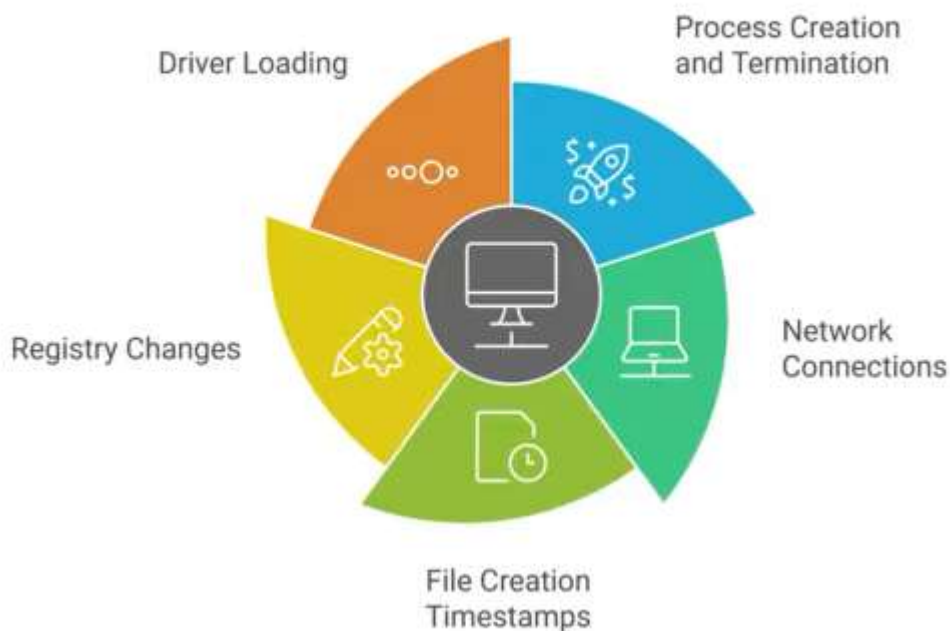


How to Install Sysmon for Advanced Windows Security Logging

1- Definition of Sysmon

Sysmon (System Monitor) is a free Windows service created by Microsoft as part of the Sysinternals suite. It runs continuously in the background to track and record key system activities—such as **process creation**, **network connections**, **file creation**, **timestamp**, **registry edits or changing**, and **driver loads**. All recorded data is saved directly to the **Windows Event Viewer**, where security teams can analyze it for threat detection and digital forensics.

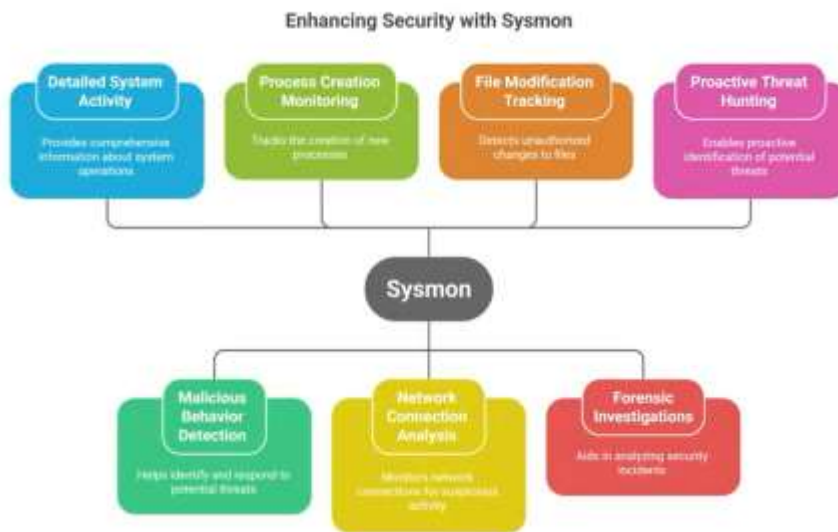
Sysmon's Comprehensive Logging Capabilities



2- The role of Sysmon in cybersecurity

Sysmon is a powerful tool for cybersecurity professionals, and it serves as a critical defense tool across four main security workflows:

- **Threat Hunting:** Flags suspicious actions, or track suspicious behavior early, such as unknown processes launching or abnormal network connections or activities.
- **Incident Response:** It helps collect forensic evidence to help investigators piece together what happened after a security breach.
- **SIEM Integration:** Forwards event logs to platforms like Splunk, Microsoft Sentinel, or the ELK Stack for real-time monitoring and alerting.
- **Malware Detection:** Catches malicious scripts and executables in action by tracking their behavior as they run.



3- How to install Sysmon on Windows agent machine

To install and use Sysmon

Go to <https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>

Download Sysmon for window → paste into C:\ → extract it (as administrator)

Go to <https://github.com/tim87tchoua/sysmon-modular>

Open sysmonconfig.xml and Download → copy and paste into C:\Sysmon.

After, open CMD as administrator → cd C:\Sysmon → run Sysmon64.exe -accepteula -i to install it.

```

C:\Windows\System32>cd C:\Sysmon
C:\Sysmon>Sysmon64.exe -accepteula -i

System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
  
```

To view all log generated do:

On search bar type event viewer → select Application and services Logs → Microsoft → Windows → Sysmon → Operational to see all the logs



How to Install Sysmon for Advanced Windows Security Logging

Step-by-Step Guide with Screenshots and Explanations
Monitor | Detect | Investigate | Protect



1 What is Sysmon?

Sysmon (System Monitor) is a free Windows service created by Microsoft as part of the Sysinternals suite. It runs continuously in the background to track and record key system activities—such as **process creation, network connections, file creation, timestamp, registry edits or changes, and driver loads**. All recorded data is saved directly to the Windows Event Viewer, where security teams can analyze it for threat detection and digital forensics.

Sysmon's Comprehensive Logging Capabilities



2 Role of Sysmon in Cybersecurity

Sysmon serves as a critical defense tool across four main security workflows:



Threat Hunting

- Flags suspicious actions, or track suspicious behavior early, such as unknown processes launching or
- abnormal network connections.



Incident Response

- Helps collect forensic evidence to help investigators piece together what happened after a security breach.



SIEM Integration

- Forwards event logs to platforms like Splunk, Microsoft Sentinel, or the ELK Stack for real-time monitoring and alerting.



Malware Detection

- Catches malicious scripts and executables in action by tracking their behavior as they run.

3 Download Sysmon

- Go to the official Microsoft Sysinternals page:

<https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>

- Click on **Download Sysmon**.

- Download Sysmon for Windows (Sysmon64.exe).



Download completed!
You will get a file named Sysmon64.exe.

4 Create a Folder and Extract Sysmon

- Create a folder (e.g., C:\Sysmon).



- Copy the downloaded Sysmon64.exe to C:\Sysmon.

- Extract it (as Administrator) if needed.



Sysmon is ready in C:\Sysmon.

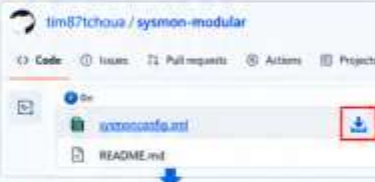
5 Download the Sysmon Configuration File

- Go to the GitHub repository:

<https://github.com/tim87tchoua/sysmon-modular>

- Download the file **sysmonconfig.xml**.

- Copy and paste it into C:\Sysmon.



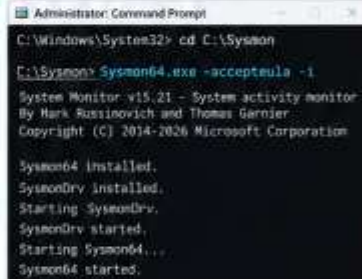
Configuration file is ready!

6 Install Sysmon

- Open Command Prompt as Administrator.

- Navigate to the Sysmon folder.

- Run the installation command:



Sysmon installed successfully!

7 Verify Sysmon Installation

- Check the Sysmon service:



- You can also check the running process:



Sysmon is running on your system.

8 View Sysmon Logs in Event Viewer

- Press **Windows + R**, type **eventvwr** and press Enter.

- Navigate to:

Applications and Services Logs → Microsoft → Windows → Sysmon → Operational



You can now see all Sysmon logs (Event ID 1, 3, 11, etc.).

9 Example: Key Event IDs

Event ID	Description
1	Process creation
3	Network connection
11	File creation
12	Registry event (create/delete)
13	Registry value set
6	Driver loaded
7	Image loaded
8	CreateRemoteThread
22	DNS query
23	Process tampering (detection)

Next Steps

- Forward Sysmon logs to your SIEM (Wazuh, Splunk, etc.)
- Create custom detection rules
- Use Sysmon for threat hunting and incident response



More Visibility
More Security



Detect Threats
Investigate Faster



Strengthen Your SOC
with Sysmon



Author: **Tims TCHOUAMOU**

Cybersecurity Enthusiast | SOC Content Creator
Learn • Practice • Secure • Share

"Knowledge
Builds a Safer
Tomorrow"